PRODUCT REQUIREMENTS DOCUMENT

LABORATORY INFORMATION MANAGEMENT SYSTEM (LIMS) — ISO/IEC 17025

MVP Ready

Nama Projek	ISO 17025 LIMS Prototype	**Versi Dokumen**	v1.0 (Final Architecture)
Sasaran Platform	Google AI Studio (Build Mode)	**Pengarang**	Product Team & System Architect
Mod Operasi	Multi-Parameter Modular	**Standard TAT**	7 Hari Bekerja (Count-down)

1. PERANAN PENGGUNA & MATRIKS KAWALAN AKSES (SEGREGATION OF DUTIES)

Bagi mematuhi syarat integriti ISO/IEC 17025, sistem menguatkuasakan pengasingan kuasa mengikut 4 peranan teras:

Peranan	Tanggungjawab Utama	Akses Modul & Hak Cipta
Admin / CS	Urus perniagaan, quotation, invoice, penerimaan bayaran & hantar laporan ke pelanggan.	Kewangan (Full), Status Bayaran, Muat Turun COA Muktamad, Penghantaran Emel/PDF.
Sample Receiving Officer (SRO)	Semak integriti sampel fizikal, pendaftaran sampel & cetakan label ID.	Borang Semak Penerimaan, Registration Form, Agihan Sampel ke Unit Makmal.
Chemist / Analyst	Terima sampel di makmal, jalankan ujian multi-parameter & isi keputusan ujian.	Lab Worklist (Assigned Tasks), Input Data Ujian, Sedia Draf COA.
Technical Manager	Semakan teknikal (QA/QC), pengesahan keputusan & kelulusan tandatangan digital.	Review & Approval Board, Audit Trail Log, Pengesahan Versi Laporan (Sign-off).

2. ALIRAN KERJA OPERASI (END-TO-END WORKFLOW)

1. **Fasa Pra-Makmal (Kewangan Wajib):** Customer Inquiry → Admin Jana Quotation (PDF) → Pengesahan Pelanggan → Admin Jana Invoice (PDF) → Bayaran Disahkan (Status: `PAID`). *(Sistem mengunci pendaftaran sampel jika bayaran belum PAID).*
2. **Penerimaan Sampel (ISO Check List):** Sampel fizikal disemak oleh SRO (Suhu, Bekas, Integriti) → Status `Accepted` / `Rejected` → Pendaftaran Sampel (Penjanaan ID Unik: `LAB-2026-XXXX`) → Agihan Ujian Modular.
3. **Analisis & Pengujian Makmal:** Analyst menerima tugasan dalam Worklist → Memasukkan keputusan ujian berasaskan *Test Method* (cth: APHA/ASTM) → Penjanaan Draf COA → Hantar untuk semakan teknikal.
4. **Kelulusan & Semakan Teknikal:** Technical Manager menyemak integriti data → Tandatangan Digital / Approve → Laporan bertukar ke status Muktamad (Final COA).
5. **Penghantaran Akhir & Penutupan TAT:** Admin memuat turun PDF muktamad → Serahkan kepada pelanggan → Pemantauan TAT dihentikan.

3. PEMATUHAN ISO 17025 & LOGIK PEMANTAUAN TAT

Ciri Pematuhan ISO 17025	Penerangan Teknikal & Syarat Operasi
Sample Condition Log	Merekodkan parameter fizikal penerimaan sampel (suhu, kerosakan, ketidakselarian borang) secara telus sebelum pendaftaran disahkan.
Audit Trail Otomatik	Log sistem merakamkan tarikh/masa, nama pengguna, dan perubahan nilai bagi setiap tindakan (Bayaran, Result Entry, Approval).
Kawalan Versi Laporan	Jika ada pindaan pada laporan muktamad, sistem menjana rujukan baharu (cth: `COA-2026-001-Rev01`) dan mengarkibkan versi lama.
Penjejak TAT (7 Hari Bekerja)	Mula: Tarikh Pendaftaran Sampel | Tamat: Tarikh Penghantaran COA ke Pelanggan. • **Hijau (Selamat)**: Baki 4 hingga 7 Hari. • **Kuning (Amaran)**: Baki 2 hingga 3 Hari. • **Merah (Kritikal/Overdue)**: Baki ≤ 1 Hari.

4. ARAHAN BINAAN GOOGLE AI STUDIO (READY-TO-COPY PROMPT)

PROMPT UNTUK SALIN KE GOOGLE AI STUDIO (BUILD MODE) **Model: Gemini 2.5 / 3 Flash**

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Act as an expert Front-end Web Developer. Build a fully functional, interactive Single Page Application (SPA) for a Laboratory Information Management System (LIMS) using React and Tailwind CSS. The system must support mock state management to demonstrate a complete user flow compliant with ISO/IEC 17025. Core Layout: 1. Sidebar Navigation: Dashboard (TAT Tracker), Finance (Quotes/Invoices), Sample Registration, Lab Analysis, Technical Review. 2. Header: Current Logged-in Role selector (Admin, SRO, Analyst, Tech Manager). The UI must restrict or show action buttons based on the active role. App Features & Flow to implement: 1. Dashboard: Show a TAT (Turn-Around Time) Tracker table. Target is 7 working days. Use color-coded badges for TAT status: Green (4-7 days left), Yellow (2-3 days left), Red (<=1 day or overdue). 2. Finance Module (Admin): Form to create a Quote, convert to Invoice, and a button to "Mark as Paid". 3. Registration (SRO): Sample registration form. CRITICAL: A sample can ONLY be registered if a linked invoice has the "Paid" status (Pre-payment mandatory condition). Include an ISO 17025 sample condition checklist (Accepted/Rejected). 4. Analysis (Analyst): A view where Analyst sees pending samples, enters multi-parameter test results (e.g., pH, BOD, Heavy Metals), and clicks "Submit Draft". 5. Approval (Tech Manager): A view to see drafts, approve them with digital sign-off, and generate a final "Certificate of Analysis". 6. Export Feature: Include a simulated "Export to PDF" button on the final Approved Report screen. Styling: Use a clean, professional, enterprise software look (whitespaces, subtle borders, rounded corners, data tables with clear borders, blue/slate color palette). Include dummy data so the tables are populated immediately upon loading. Provide the complete code.
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